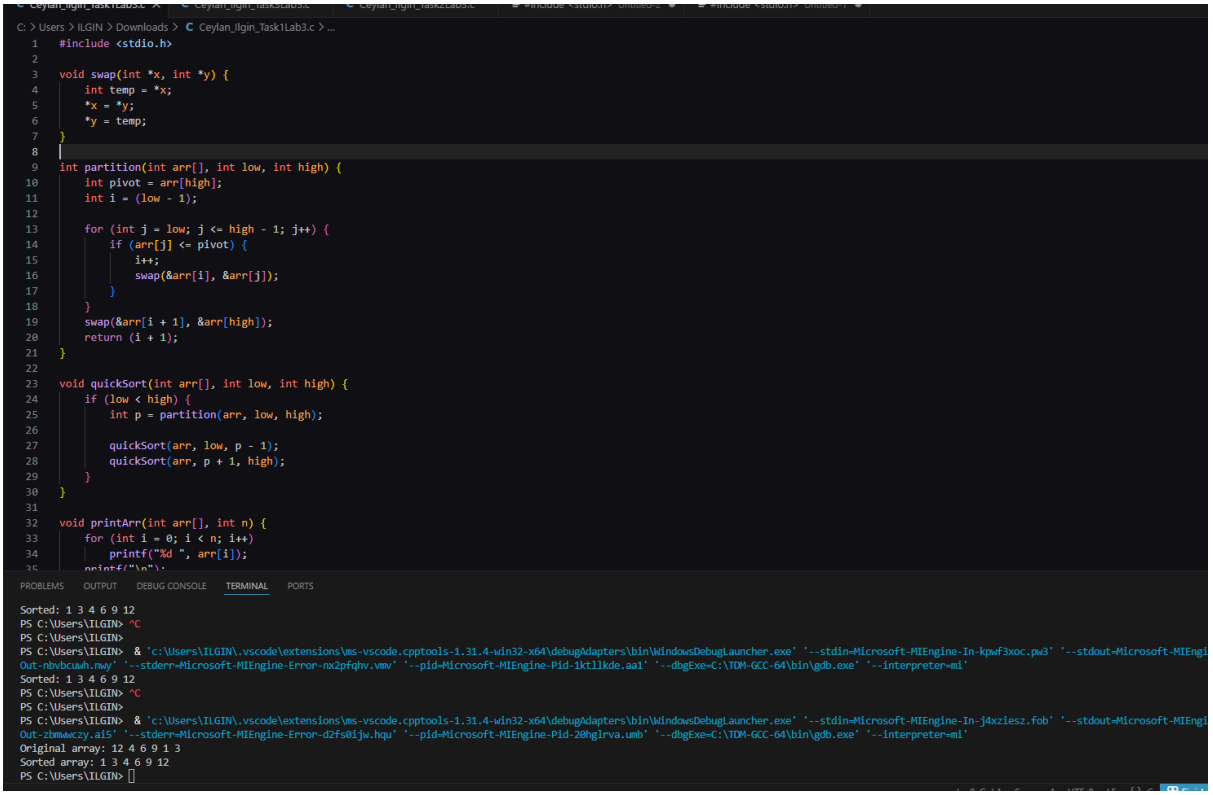
Algorithms and Data Structures

REPORT LAB 3

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TASK1



```c
#include <stdio.h>

void swap(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = (low - 1);

    for (int j = low; j <= high - 1; j++) {
        if (arr[j] <= pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int p = partition(arr, low, high);

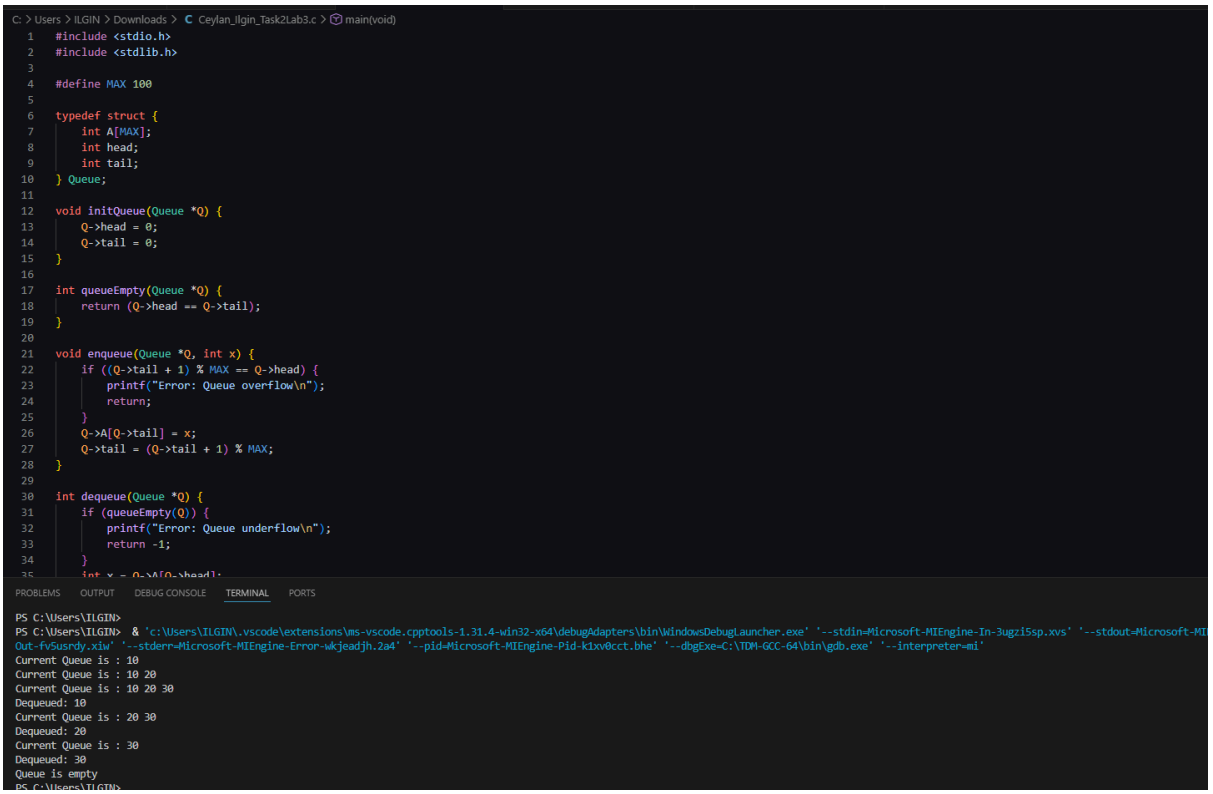
        quickSort(arr, low, p - 1);
        quickSort(arr, p + 1, high);
    }
}

void printArr(int arr[], int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
```

```
Sorted: 1 3 4 6 9 12
PS C:\Users\ILGIN> ^C
PS C:\Users\ILGIN>
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-kpwf3xoc.pw3' '--stdout=Microsoft-MIEngi
Out-nbvbcuwh.nwy' '--stderr=Microsoft-MIEngine-Error-nx2pfqhv.vmv' '--pid=Microsoft-MIEngine-Pid-1ktllkde.aa1' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Sorted: 1 3 4 6 9 12
PS C:\Users\ILGIN> ^C
PS C:\Users\ILGIN>
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-j4xziesz.fob' '--stdout=Microsoft-MIEngi
Out-zbmwwczy.ai5' '--stderr=Microsoft-MIEngine-Error-d2fs0ijw.hqu' '--pid=Microsoft-MIEngine-Pid-20hglrva.umb' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Original array: 12 4 6 9 1 3
Sorted array: 1 3 4 6 9 12
PS C:\Users\ILGIN>
```

-Sorted: 1 3 4 6 9 12

TASK2



```c
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

typedef struct {
    int A[MAX];
    int head;
    int tail;
} Queue;

void initQueue(Queue *Q) {
    Q->head = 0;
    Q->tail = 0;
}

int queueEmpty(Queue *Q) {
    return (Q->head == Q->tail);
}

void enqueue(Queue *Q, int x) {
    if ((Q->tail + 1) % MAX == Q->head) {
        printf("Error: Queue overflow\n");
        return;
    }
    Q->A[Q->tail] = x;
    Q->tail = (Q->tail + 1) % MAX;
}

int dequeue(Queue *Q) {
    if (queueEmpty(Q)) {
        printf("Error: Queue underflow\n");
        return -1;
    }
```

```
PS C:\Users\ILGIN>
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-3ugzi5sp.xvs' '--stdout=Microsoft-MI
Out-fv5usrdy.xiw' '--stderr=Microsoft-MIEngine-Error-wkjeadjh.2a4' '--pid=Microsoft-MIEngine-Pid-k1xv0cct.bhe' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Current Queue is : 10
Current Queue is : 10 20
Current Queue is : 10 20 30
Dequeued: 10
Current Queue is : 20 30
Dequeued: 20
Current Queue is : 30
Dequeued: 30
Queue is empty
PS C:\Users\ILGIN>
```

TASK3

C: > Users > ILGIN > Downloads > C Ceylan_Ilgin_Task3Lab3.c > ...

```c
#include <stdio.h>
#include <stdlib.h>

struct node {
    int key;
    struct node* prev;
    struct node* next;
};

struct node* createNode(int key) {
    struct node* newNode = (struct node*)malloc(sizeof(struct node));
    if (newNode == NULL) {
        printf("Memory allocation failed!\n");
        exit(1);
    }
    newNode->key = key;
    newNode->prev = NULL;
    newNode->next = NULL;
    return newNode;
}

struct node* insertBegin(struct node* head, int key) {
    struct node* x = createNode(key);
    x->next = head;
    if (head != NULL) {
        head->prev = x;
    }
    return x;
}

struct node* insertEnd(struct node* head, int key) {
    struct node* x = createNode(key);
    if (head == NULL) {
        return x;
    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ILGIN> ^C
PS C:\Users\ILGIN>
PS C:\Users\ILGIN>  & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-4t32xti3.xzq' '--stdout=Microsoft-MIEngine-Out-vw00plry.itc' '--stderr=Microsoft-MIEngine-Error-ut4wmush.hxc' '--pid=Microsoft-MIEngine-Pid-h2nofroq.rqu' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
List (forward): 5
List (forward): 20 5
List (forward): 20 5 40
List (forward): 20 5 40 60
Found: 20
List (forward): 20 80 5 40 60
Found: 40
List (forward): 20 80 5 60
```